Nieb Hasan Neom

Rolla, MO
niebhasanneom0@gmail.com
linkedin.com/in/nieb-hasan-neom github.com/NiebHasan077 Google Scholar ORCID

Summary

Computer Science Ph.D. student and **Kummer Doctoral Fellow** at Missouri S&T researching **agentic optimization of large-scale scientific data movement**, with LLM agents that reach **94–98%** of the best fixed transfer setting without prior tuning. **Four peer-reviewed publications**, **266 citations**, including first-author work at **WORKS26 @ SC'26** and a co-authored paper presented at **IEEE eScience 2026**, alongside three years of backend engineering on a 30-microservice platform.

Education

Missouri University of Science & Technology **Rolla, MO, USA**
Doctor of Philosophy in Computer Science *Aug. 2025 – Present*

- **Kummer Innovation and Entrepreneurship Doctoral Fellow** | **GPA: 4.00/4.00** | Anticipated graduation: **12/2029**

Khulna University of Engineering & Technology **Khulna, Bangladesh**
Bachelor of Science in Computer Science and Engineering *Feb. 2017 – April 2022*

- **CGPA: 3.51/4.00** | KUET Technical Scholarship (2017–2022)

Research Experience

Computer Systems & Networking Lab, Missouri University of Science & Technology **Rolla, MO**
Graduate Research Assistant | Advisor: Dr. Md Arifuzzaman *Aug. 2025 – Present*

- Led **NetBench** (GitHub; paper under review), a study of how LLMs acquire high-performance networking knowledge: built **HPN-QA**, a 233-question audited benchmark, from a 2,371-paper corpus, and adapted **8 open-weight models (1B–27B)** via continued pre-training, full and LoRA fine-tuning, and retrieval with **PyTorch, HuggingFace Transformers, PEFT and DeepSpeed** on H100, V100 and A6000 GPUs
- Showed that retrieval improved **every model** and brought 9B and 12B local models to scores **statistically equivalent to closed-book Claude Sonnet 4.6 and Gemini 2.5 Pro**, while full fine-tuning *reduced* the largest models' scores; validated the LLM judge against two human experts (Pearson $r = 0.89$)
- Built an **Agentic Data Transfer Optimizer** in which local LLM diagnostic and decision agents tune transfer concurrency each round from TCP, host and throughput telemetry; on CloudLab (10 Gbps) and an InfiniBand HPC cluster it reached **94–98%** of the best fixed setting's throughput objective **without knowing that setting in advance**, averaging up to **84.5 Gbps**
- Designed **guarded, state-aware orchestration** extending LLM function calling with workflow semantics—multi-turn input accumulation, typed-schema argument constraints and blocked execution on incomplete state—giving domain scientists natural-language access to trained prediction models (**first author, accepted at WORKS26 @ SC'26**)
- Instantiated the workflow for protein-corona prediction over a backend of **178 per-protein classifiers (87.13% AUC-ROC)**; measured **98.5% feature-extraction accuracy** across 300 cases and **300/300** passing reliability scenarios spanning 50 nanoparticle configurations, in collaboration with Louisiana State University
- Co-authored **FastBioDL** (**IEEE eScience 2026**, presented), an adaptive segmented downloader for public genomic repositories whose online worker controller led throughput in **8 of 9** workload–testbed combinations on SDSC Expanse, FABRIC and a lab server, reaching up to **3.88**× the best genomic-download baseline and **1.91**× aria2c

Industry Experience

Kona Software Lab Limited **Gulshan, Dhaka, Bangladesh**
Software Engineer II (promoted from Graduate Engineer, 2024) *Apr. 2022 – Jun. 2025*

- Backend engineer on **Kona Token Trade**, a cryptocurrency and blockchain-based token trading platform; owned **6** of the platform's **30 Spring Boot microservices** and contributed to **15** in total, covering trade, wallet, user, payment and fee management
- Replaced synchronous inter-service REST calls with **Kafka** messaging, closing a fault-tolerance gap in which a single unavailable downstream service could fail an in-flight transaction; introduced caching that cut system response time by **30%**

- Implemented **two-factor authentication** on critical financial operations such as asset buy and sell, and integrated bank-account and KONA card payment gateways
- Redeployed parent company Kona I's **Card Personalization System** (Apache Struts, Java) and configured profiles for different customers; it is in production at **Islami Bank Bangladesh**, the country's largest private bank, personalizing its VISA debit cards. Selected from the engineering team for a **14-day training program** at headquarters in **Seoul, South Korea** on the system and the **GlobalPlatform, Visa, MasterCard and EMV** specifications
- Built against **AWS** (S3, SNS, EC2) from the backend services; mentored and onboarded **2 junior engineers** into backend engineering fundamentals

Publications

- **Nieb Hasan Neom**, Manas Ranjan Gartia, and Md Arifuzzaman. "Guarded, State-Aware Orchestration for Schema-Constrained Scientific Model Execution." *21st Workshop on Workflows in Support of Large-Scale Science (WORKS26)*, held in conjunction with *SC26*, Chicago, IL, Nov. 2026. **(Accepted — to appear)**
- Rasman Mubtasim Swargo, **Nieb Hasan Neom**, and Md Arifuzzaman. "Self-Tuning Segmented Retrieval of Large-Scale Genomic Data." *22nd IEEE International Conference on e-Science (eScience 2026)*, Naples, Italy, Oct. 2026. **Short paper; presented.**
- Sunanda Das, Md Samir Imtiaz, **Nieb Hasan Neom**, Nazmul Siddique, and Hui Wang. "A hybrid approach for Bangla sign language recognition using deep transfer learning model with random forest classifier." *Expert Systems with Applications* 213 (2023): 118914. **Impact Factor: 8.665 | 218 citations. Link**
- Md Milon Islam, **Nieb Neom**, Md Samir Imtiaz, Sheikh Nooruddin, Md Repon Islam, and Md Rabiul Islam. "A Review on Fall Detection Systems Using Data from Smartphone Sensors." *Ingénierie des Systèmes d'Information* 24, no. 6 (2019): 569–576. **48 citations. Link**

Technical Skills

Languages: Python, Java, C, C++, SQL, Bash
Machine Learning & LLMs: LLM fine-tuning (full, LoRA), continued pre-training, retrieval-augmented generation, LLM-as-judge evaluation, function calling and agentic orchestration; PyTorch, TensorFlow, HuggingFace Transformers, PEFT, DeepSpeed, Ollama, OpenAI and Gemini APIs
HPC & Systems: CloudLab, FABRIC, TACC/ACCESS, Falcon data mover, TCP performance analysis, multiprocessing, zero-copy `sendfile()`, Linux systems programming
Distributed Systems & Backend: Spring Boot, Apache Kafka, microservice architecture, RESTful web services, AWS (S3, SNS, EC2)
Databases: PostgreSQL, MongoDB, Redis, MySQL
Developer Tools: Git, Gradle, Jenkins, Jira, IntelliJ IDEA, DataGrip, VS Code, LaTeX

Honors & Awards

- **Kummer Innovation and Entrepreneurship Doctoral Fellowship**, Missouri S&T (2025–present); **1st Runner-up**, KonaSL Coding Combat 2023; **Best Performer (2016 batch)** and 11th of 90, Intra-KUET Programming Contest 2020; Rank **10 of 73** teams, Intra-KUET Programming Contest 2019
- **Codeforces**: **Expert**, max rating **1629**, 550+ problems over 145+ contests | **LeetCode**: **Knight**, max rating **1895** (**global top 5%**), 210+ problems | **CodeChef**: 4-star, max rating 1819 | **UVa**: 140+ | **1100+** problems solved overall
- **Fine-tuning & RL for LLMs: Intro to Post-training**, DeepLearning.AI (2026); Deep Learning Specialization, Coursera

Leadership, Service & Community

- **Volunteer**, ICRM Distribution Center, Islamic Center of Rolla, Missouri (Sept. 2025 – Present) — weekly halal meat distribution to community members and monthly facility cleaning
- **Student Volunteer**, SC26 — selected, with registration, airfare and accommodation covered; **Reviewer**, *Journal of Autonomous Intelligence* (Frontier Scientific Publishing)
- **Executive Member**, SGIPC — Special Group Interested in Programming Contest, KUET (2018–2022) — ran competitive-programming onboarding workshops for 60+ first-year students and arranged online and onsite contests
- **Assistant Organizing Secretary**, KUET Math Club (2020–2022) — organized the departmental math festival and ran an olympiad training workshop for 30+ participants
- **Executive Member**, Organization of KUET Sports (2020–2022) — organized annual sporting events; captained the CSE departmental cricket team